

Multiple Choice (10 marks)

1. If 1.0 g of rubidium and 1.0 g of bromine are reacted, what will be left in measurable amounts (more than 0.10 mg) in the reaction vessel?
 - (a) RbBr only
 - (b) RbBr, Rb, Br₂, and Rb₂Br
 - (c) RbBr and Rb₂Br only
 - (d) RbBr, Rb, and Br₂
 - (e) Rb and Br₂ only
2. Cerium(III) sulfate, Ce₂(SO₄)₃, is less soluble in hot water than it is in cold. Which of the following conclusions may be related to this?
 - (a) The hydration energies of cerium ions and sulfate ions are very high.
 - (b) The heat of solution of cerium(III) sulfate is isothermic.
 - (c) The solubility of cerium(III) sulfate increases with temperature.
 - (d) The heat of solution of cerium(III) sulfate is exothermic.
 - (e) The hydration energies of cerium ions and sulfate ions are very low.
3. Calculate the ratio of the electrical and gravitational forces between a proton and an electron.
 - (a) 4×10^{39}
 - (b) 6×10^{36}
 - (c) 7×10^{37}
 - (d) 5×10^{39}
 - (e) 9×10^{41}
4. When 10.0 g of n-propylchloride is allowed to react with excess sodium in the Wurtz reaction, how many grams of hexane would be produced assuming a 70% yield?
 - (a) 8.67 g
 - (b) 3.82 g
 - (c) 5.11 g
 - (d) 3.00 g
 - (e) 4.45 g
5. What is the molar mass of a monoprotic weak acid that requires 26.3 mL of 0.122 M KOH to neutralize 0.682 gram of the acid?

- (a) 26.3 g mol⁻¹
- (b) 500 g mol⁻¹
- (c) 212 g mol⁻¹
- (d) 0.122 g mol⁻¹
- (e) 1.22 g mol⁻¹

True / False (10 marks)

Answer True or False

6. True or False: The van der Waals parameter a is measured in SI base units as $7.61 \times 10^1 - 2 \} kg m^5 s^{-2} mol^{-2}$.
6. True or False: One liter of paint, brushed out to a thickness of 100 microns, will cover an area of 30 square meters.
7. True or False: The mass of water vapor present in a $400 m^3$ room at $27^\circ C$ and 60% relative humidity is 6.2 kg.
7. True or False: The thermal stability of group-14 hydrides decreases in the order $CH_4 < SiH_4 < GeH_4 < SnH_4 < PbH_4$.
8. True or False: Sulfur dioxide (SO₂) is classified as an acid anhydride due to its ability to form sulfurous acid upon hydration.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the frequency of a light wave that corresponds to the wavelength equal to the Bohr radius ($0.5292 \times 10^{-8} cm$). Discuss the implications of this relationship in quantum mechanics.
11. Discuss the criteria for Hermitian operators in quantum mechanics and analyze why the operators d/dx and d^2/dx^2 qualify as Hermitian, including their implications in chemical systems.

End of Paper